Faculty of Technology and Engineering

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Practical Performa

Academic Year	:	2025-26	Semester	:	3rd
Course code	:	CSUC201	Course name	:	Fundamentals of Data Structure and Algorithms

Practical- 7 Queue

Questions and Answers (Handwritten):

7.1 Match the queue types in List–I with their corresponding properties/applications in List–II:

List – I (Queue Concept) | List – II (Property / Application)
(a) Simple Queue
(b) Circular Queue
(c) Double-Ended Queue (Deque)
(d) Priority Queue
(e) Queue using Linked List

(i) Elements can be inserted/deleted from both ends.
(ii) Overcomes the wasted space problem of linear queues using wrap-around.
(iii) Insert at rear, delete from front; implemented using arrays or linked lists.
(iv) Each element has a priority; higher priority elements are dequeued first.
(v) Provides dynamic memory allocation, avoiding fixed-size limitations.

Answer Codes:
(a) → ___
(b) → ___
(c) → ___
(d) → ___
(e) → ___

7.2 Consider the following C program:

```
#include
#define EOF -1
    void push (int); /* push the argument on the stack */
    int pop (void); /* pop the top of the stack */
    void flagError ();
    int main () {
      int c, m, n, r;
      while ((c = getchar ()) != EOF) {
        if (isdigit (c) )
          push (c);
        else if ((c == '+') || (c == '*')) {
          m = pop ();
          n = pop ();
          r = (c == '+') ? n + m : n*m;
          push (r);
        } else if (c != ' ')
          flagError ();
      }
      printf("% c", pop ());
    }
```

What is the output of the program for the following input ? 5 2 * 3 3 2 + * +

Please Justify.

- **A. 15**
- **B. 25**
- **C. 30**
- **D. 150**

7.3 (Circular Queue Operations)

We use a circular queue of size 5 (implemented using array). Perform the following operations step by step:

Insert(10), Insert(20), Insert(30), Insert(40), Delete(), Insert(50), Insert(60), Delete(), Insert(70)

For this sequence:

(a) Show the queue contents after each operation.

(b) Indicate the position of front and rear pointers after every step.

(c) Explain how circular queue solves the problem of "false overflow" in simple queues.

(d) State the time complexity of Insert and Delete operations in circular queue.

7.4 Consider a sequence aa of elements $a_0=1, a_1=5, a_2=7, a_3=8, a_4=9$, and $a_5=2$. The following operations are performed on a stack S and a queue Q, both of which are initially empty.

I: push the elements of a from a_0 to a_5 in that order into S.

II: enqueue the elements of a from a_0 to a_5 in that order into Q.

III: pop an element from S.

IV: dequeue an element from Q.

V: pop an element from S.

VI: dequeue an element from Q.

VII: dequeue an element from Q and push the same element into S.

VIII: Repeat operation VII three times.

IX: pop an element from S.

X: pop an element from S.

The top element of S after executing the above operations is ___.

Grade/Marks **Sign of Lab Teacher with Date**

(_____ / 10)